Student Academic Performance Prediction

Machine Learning Project Report

Project Area	Supervised Machine Learning
Dataset	UCI Student Performance — Mathematics
Regression Models	Linear Regression, Decision Tree, Random Forest
Classification Model	Random Forest Classifier
Main Target	Final Grade (G3), 0–20

Prepared as a practical project demonstrating data analysis, machine learning, model evaluation and interpretation.

Dataset records used: 395 students

1. Project Overview

This project predicts student academic performance using the Student Performance dataset from the UCI Machine Learning Repository. The analysis uses the Mathematics dataset (student-mat.csv), which contains demographic, social, school-related and academic information.

The main task is regression: predicting the final Mathematics grade (G3). Three models are compared—Linear Regression, Decision Tree Regression and Random Forest Regression. A second task converts G3 into a Pass/Fail outcome and uses a Random Forest Classifier.

The analysis uses an 80/20 train-test split with random_state = 42. Categorical variables are one-hot encoded before modeling.

Best regression model in this run: Random Forest. It achieved an R^2 of 0.805 on the held-out test set.

2. Objectives

- Understand the structure and characteristics of student performance data.
- Explore relationships between study habits, attendance, previous grades and final academic performance.
- Prepare categorical and numerical variables for machine learning.
- Compare Linear Regression, Decision Tree Regression and Random Forest Regression.
- Evaluate regression models using MAE, RMSE and R^2.
- Build a Pass/Fail classification model using Random Forest.
- Evaluate classification using accuracy, precision, recall, F1-score and ROC-AUC.
- Use visualizations and feature importance to interpret the results.

3. Dataset

The Student Performance dataset was created from student achievement data collected at two Portuguese schools. The UCI documentation describes 649 Mathematics records and 33 attributes in the current repository metadata. The dataset supports both regression and classification tasks. The final grade G3 is the regression target.

Item	Details
Source	UCI Machine Learning Repository
File	student-mat.csv
Records	649
Target	G3 — final grade (0–20)
Earlier grades	G1 — first period; G2 — second period
Feature groups	Demographic, family, school, study, social and academic variables
Missing values	None reported in the UCI dataset documentation
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Modeling consideration: G1 and G2 are earlier grades and are strongly related to G3. This makes the final-grade prediction easier, but a separate early-warning model should exclude G1 and G2 if those grades are not yet available.

4. Methodology

The project follows these steps:

- Load the Mathematics dataset using the semicolon delimiter.
- Check data types, missing values, duplicates and descriptive statistics.
- Create the Pass target using G3 ≥ 10.
- Separate predictors and target variables.
- One-hot encode categorical features.
- Split data into training and testing sets.
- Train the three regression models.
- Train a Random Forest classifier for Pass/Fail.
- Evaluate the models on the test data.
- Create plots for interpretation and model comparison.

Model settings used in this report

Model	Main setting
Linear Regression	Standard linear regression
Decision Tree	max_depth = 5, random_state = 42
Random Forest Regression	200 trees, random_state = 42
Random Forest Classification	200 trees, random_state = 42
Train/Test Split	80% training / 20% testing

5. Exploratory Data Analysis

The following graphs are generated directly from the dataset used for this report.

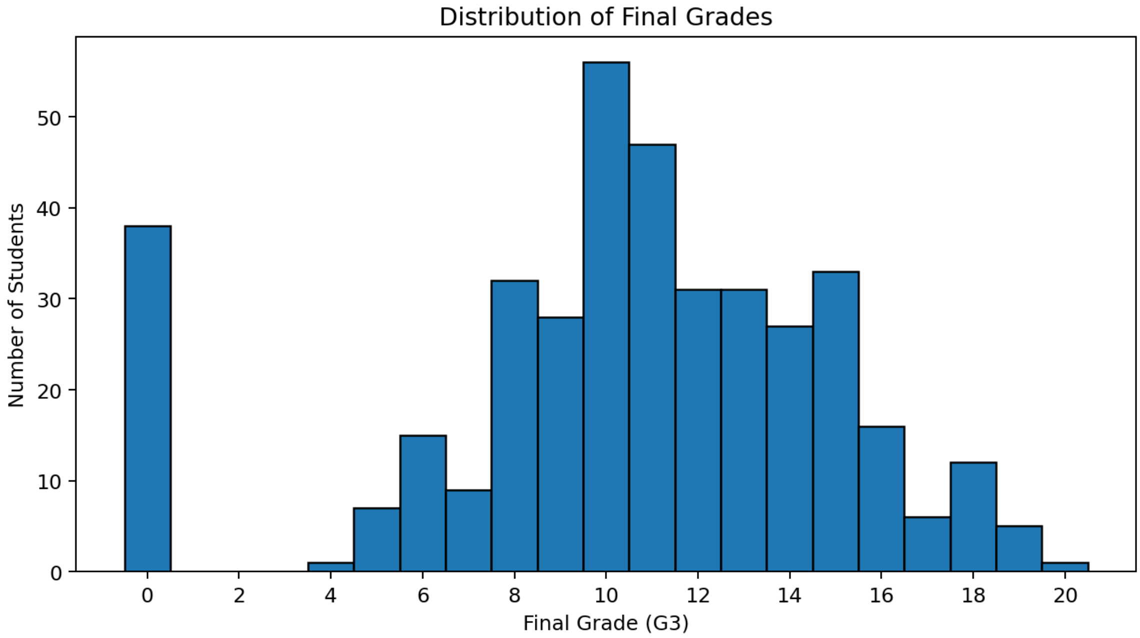

Figure 1. Distribution of final grades (G3).

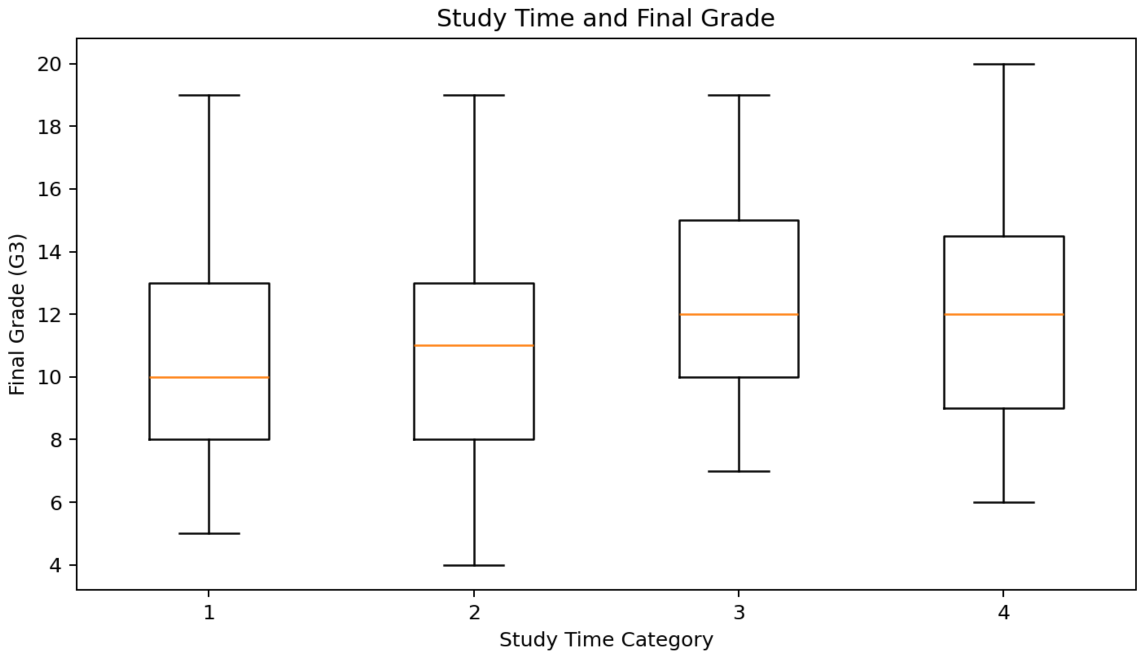

Figure 2. Final-grade distribution across study-time categories.

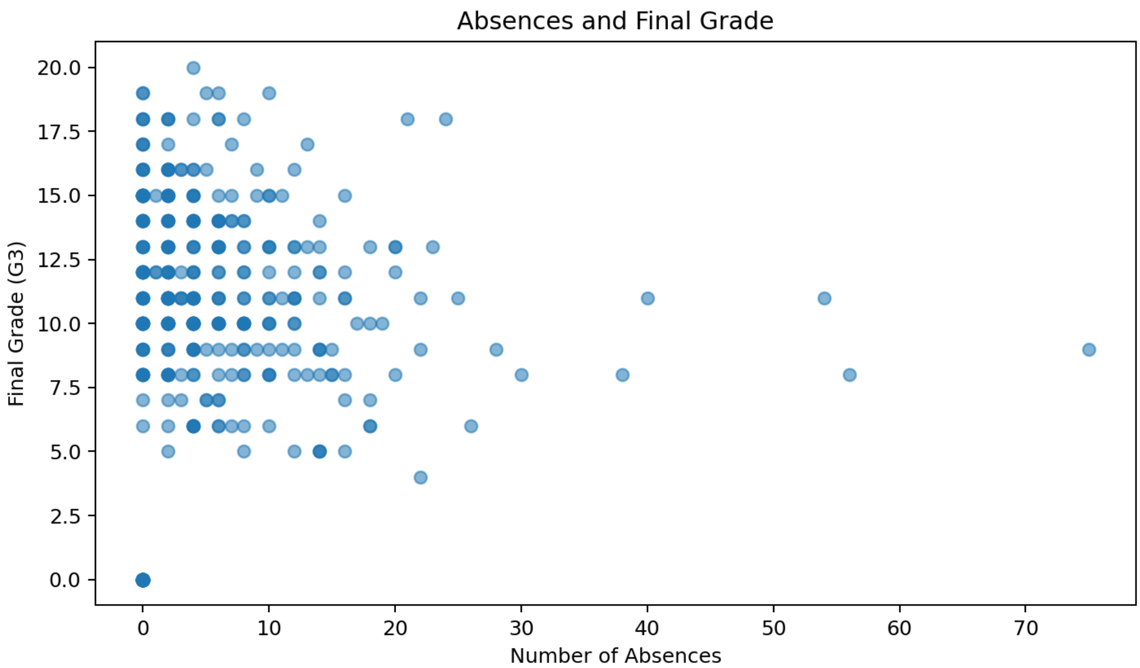

Figure 3. Relationship between absences and final grade.

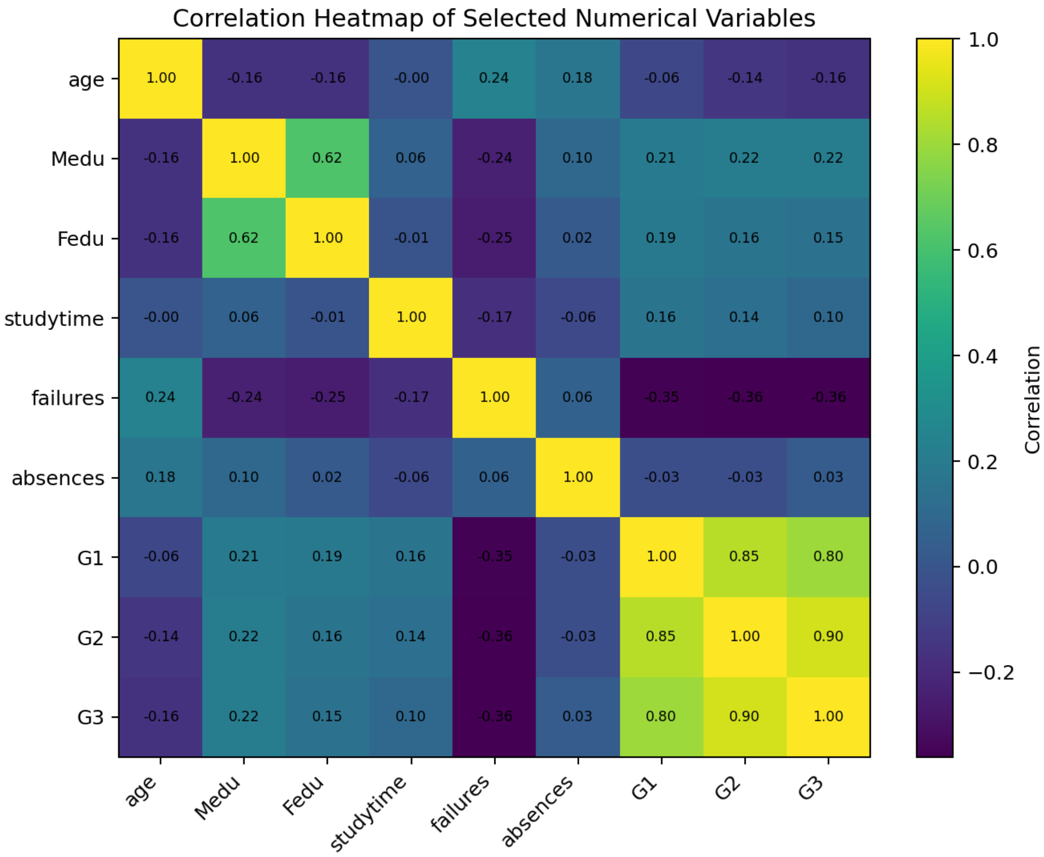

Figure 4. Correlations among selected numerical variables.

6. Regression Results

The three regression models were evaluated on the same 20% test set. Lower MAE and RMSE indicate smaller prediction errors, while a higher R^2 indicates that more variance in G3 is explained.

Model	MAE	RMSE	R^2
Linear Regression	1.647	2.378	0.724
Decision Tree	1.341	2.404	0.718
Random Forest	1.178	2.001	0.805

The best R^2 in this experiment was obtained by **Random Forest**. The actual-vs-predicted plot below provides a visual check of prediction quality.

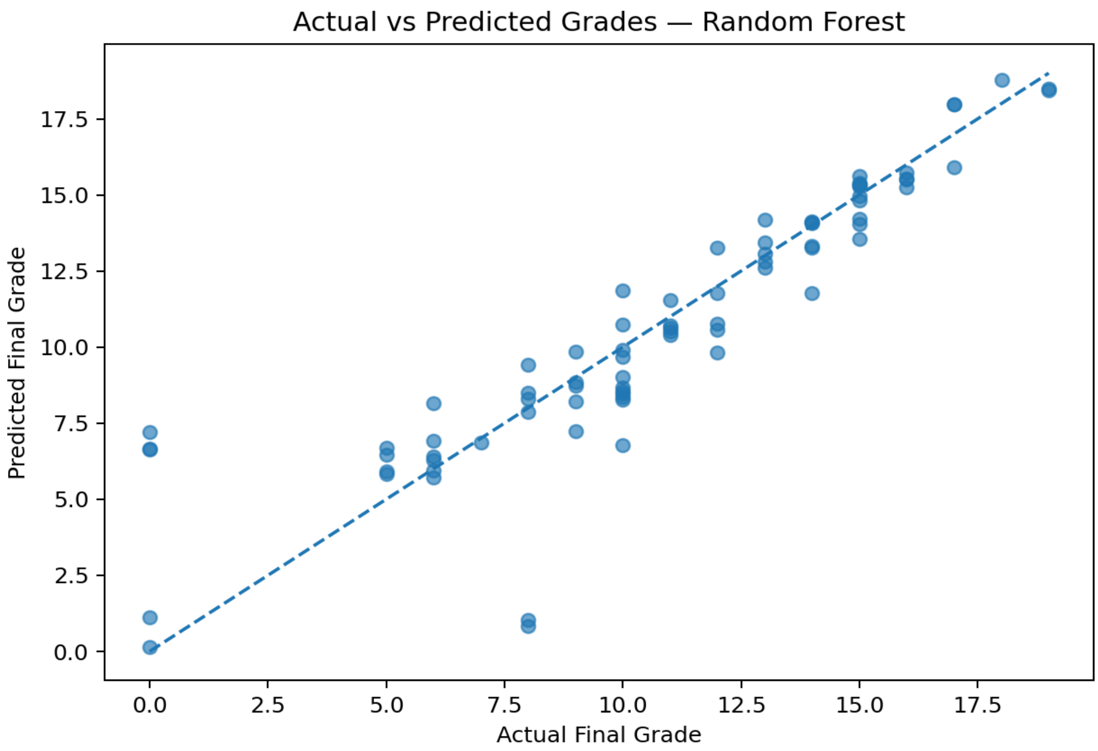

Figure 5. Actual versus predicted final grades for the best-performing regression model (Random Forest).

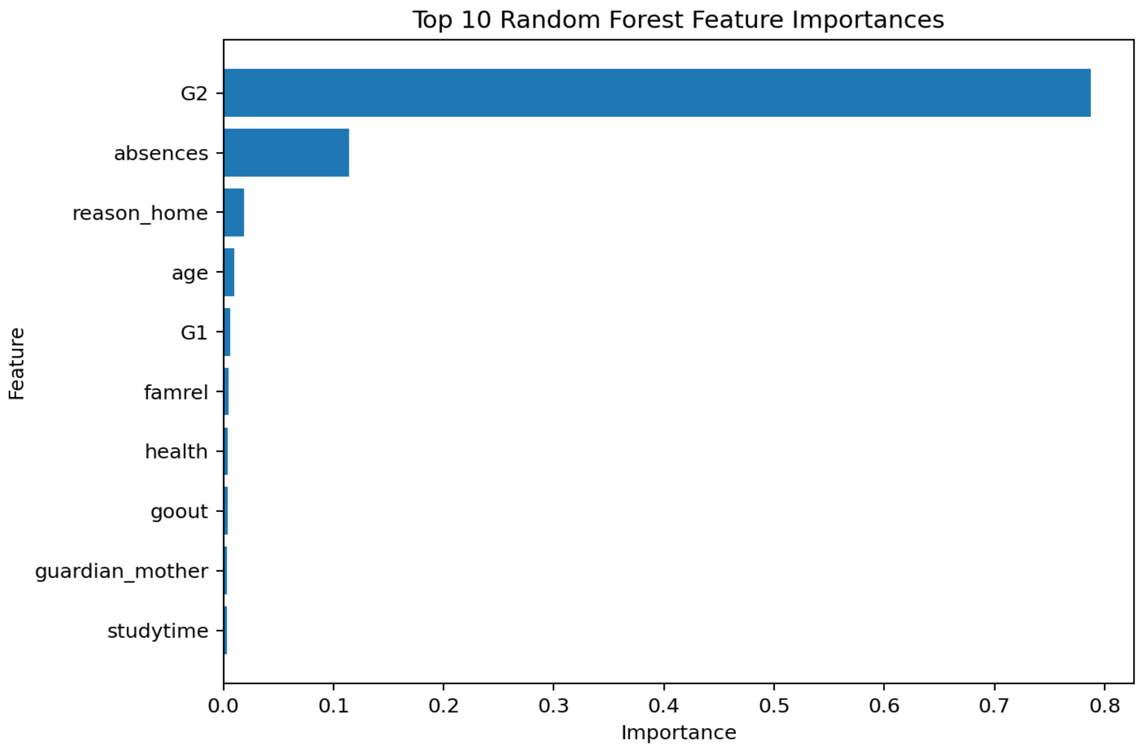

Figure 6. Top 10 feature importances from the Random Forest regression model.

7. Pass/Fail Classification

For the secondary task, students with G3 ≥ 10 are labeled Pass and students with G3 < 10 are labeled Fail. A Random Forest Classifier with 200 trees is trained using the same predictor set.

Metric	Result
Accuracy	0.873
Precision	0.957
Recall	0.849
F1-score	0.900
ROC-AUC	0.924

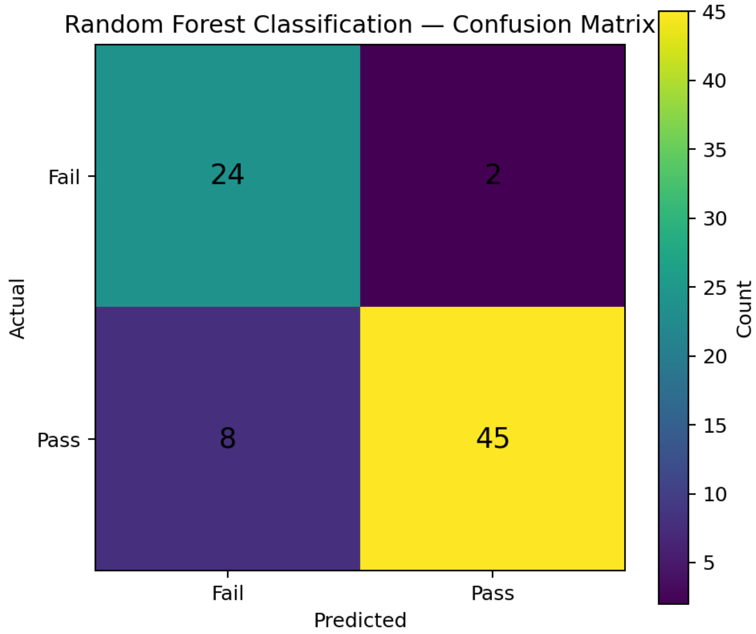

Figure 7. Confusion matrix for Pass/Fail classification.

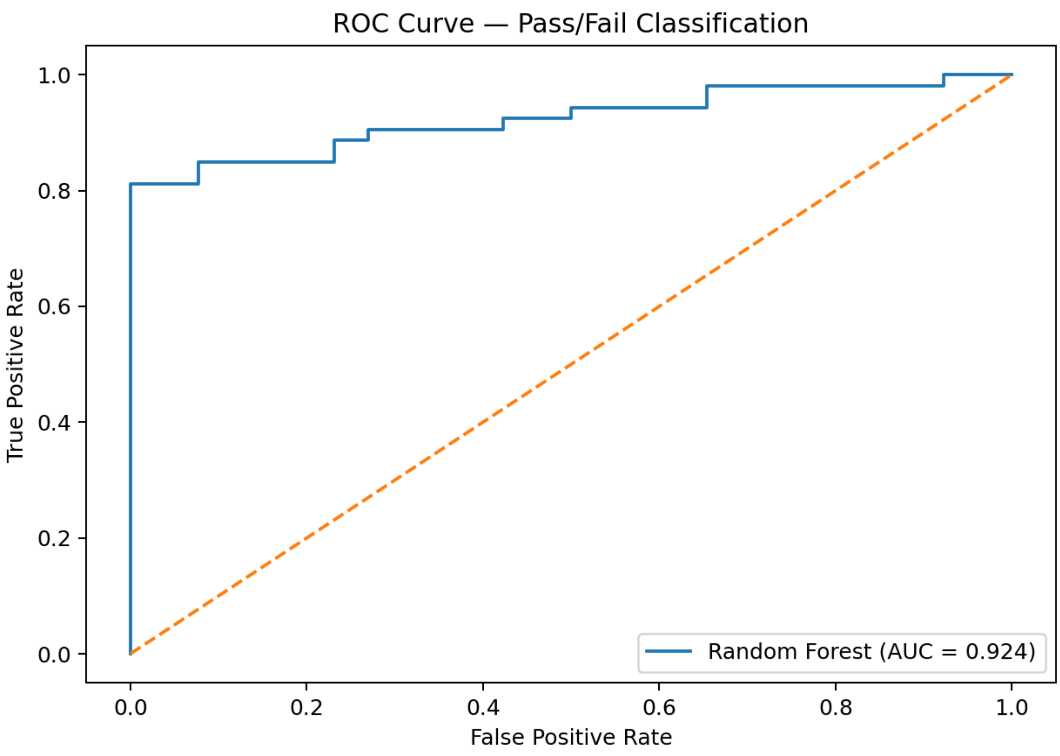

Figure 8. ROC curve for the Random Forest Pass/Fail classifier.

8. Discussion

The regression comparison shows how a simple linear model compares with tree-based approaches on the same student-performance data. Tree-based models can represent non-linear relationships and interactions, while Linear Regression provides a useful baseline.

The feature-importance graph should be interpreted as predictive importance rather than causation. In particular, G1 and G2 are expected to be highly informative because they are earlier grades for the same course.

The classification results should be read using several metrics together. Accuracy alone can hide differences between false positives and false negatives, so precision, recall, F1-score and ROC-AUC provide additional context.

9. Limitations

- The dataset represents students from two Portuguese schools and may not generalize to other educational settings.
- G1 and G2 provide strong information about G3; excluding them would create a more difficult but potentially more useful early-warning scenario.
- A single train-test split provides one estimate of performance; cross-validation could provide a more stable comparison.
- Feature importance does not establish causal relationships.
- Student predictions should be used as decision-support information rather than as definitive judgments about individual students.

10. Conclusion

This project demonstrates an end-to-end machine learning workflow for student academic performance prediction. The analysis combines data exploration, preprocessing, regression, classification, evaluation and visualization.